



Dynamic overcharge investigations of lithium ion batteries with different state of health

Lei Feng, Lihua Jiang, Jialong Liu, Zhaoyu Wang, Zesen Wei, Qingsong Wang^{*}

State Key Laboratory of Fire Science, University of Science and Technology of China, Hefei, 230026, China

HIGHLIGHTS

- The influence of SOH on the thermal runaway behavior is investigated.
- The safety of the batteries during overcharge decreases with the decrease of SOH.
- The contribution of side reactions to thermal runaway does not depend on the SOH.
- A feasible warning measure for impending risk of thermal runaway is provided.

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ABSTRACT

Overcharge and even further thermal runaway of lithium ion batteries may occur when there are inconsistencies between batteries, charging devices or battery management system fails. In this work, the thermal behavior and heat accumulation of commercial lithium ion batteries with different states of health (SOH) are studied for aging batteries when they are subjected to overcharging. The results show that the thermal runaway is triggered by the melting of separator which leads to internal short circuit, and this triggering factor does not change with SOH. Through the changes of temperature, voltage and duration before the thermal runaway, the safety decreases after aging, which is mainly due to the loss of lithium ions and the change of negative capacity after the battery is overcharged at a high rate. The contribution rate of heat generated by side reaction to thermal runaway is almost not affected by SOH. In addition, overcharge experiments of batteries with 80% SOH are carried out under different current rates. The results provide a method for warning the risk of thermal runaway. When the voltage reaches an inflection point, safety measures should be taken within 3 min.

1. Introduction

Lithium ion batteries have the characteristics of high energy density, low self-discharge rate, low memory effect and so on. It dominates the current rechargeable market and has been commercialized for the first time by Sony Energy Tech [1]. Lithium ion batteries have been widely used in portable electronic products such as digital cameras and notebook computers, as well as electric transportation such as electric vehicles (EVs) and hybrid electric vehicles (HEVs) [2,3], while the safety problem during its application has attracted more and more attention in recent years, especially when used under abuse conditions. Overcharge is one of the common conditions of abuse, under which lithium ion batteries may burst, release smoke, catch fire or even explode [4–8]. Overcharge can easily occur due to inconsistencies between the

batteries, especially if the charger fails or the battery management system (BMS) is not designed properly [9–12]. In addition, due to a large number of retired lithium ion batteries, the way to dispose these batteries become an important issue. Many researchers have studied the feasibility and economy of the secondary use of lithium ion batteries [13–18]. However, there are still many problems in the use of aging batteries, such as their safety issues [18].

Much research has been carried out on improving the overcharge performance of lithium-ion batteries, such as adding anti-overcharge electrolyte additive [19,20]. Nevertheless, thermal runaway caused by overcharge is still very easy to appear, which causes serious safety accidents. There have been a lot of studies on the safety of batteries under overcharge [21]. When the battery continues to be charged beyond the cut-off voltage, the structure of positive electrode will undergo changes,

^{*} Corresponding author.

E-mail address: pinew@ustc.edu.cn (Q. Wang).

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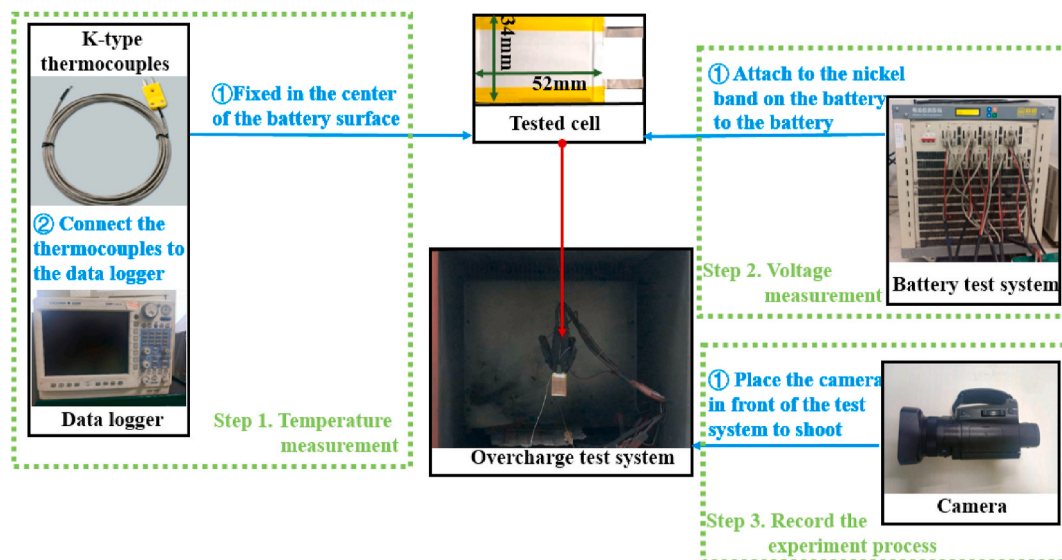


Fig. 1. Schematic diagram of experimental set-up of overcharge tests.

which will increase the impedance and generate a large amount of joule heat. At the same time, oxygen will be released and the oxidation of electrolyte will generate a lot of heat. For the negative electrode, a large amount of lithium metal will be deposited on the surface. On the one hand, it will react with the electrolyte to increase the thickness of SEI film, leading to the increase of ohmic impedance and generating a large amount of Joule heat, on the other hand, it may puncture the separator and cause an internal short circuit. The rapid accumulation of heat and gas will cause the battery to expand, and eventually lead to thermal runaway [21,22]. Ren et al. [23] build an electrochemical-thermal coupled overcharge-to-thermal-runaway (TR) model. The mechanism of thermal runaway caused by overcharge is well explained by this model, in addition, the key parameters were further modeled and analyzed, and the results show that the two main heat sources leading to thermal runaway are the exothermic oxygen of electrolyte and the reaction between lithium deposition and electrolyte during overcharging. The process of overcharging a large lithium ion battery with $\text{Li}_{1/3}\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2 + \text{Li}_y\text{Mn}_2\text{O}_4$ composite cathode is divided into four detailed stages by Ouyang et al. [24], the first stage is that before the battery is overcharged to 120%SOC, there is no obvious capacity degradation behavior of the battery. In the second stage, the battery continues to charge to 140%SOC, resulting in lithium deposition, continuous thickening of SEI film, loss of cathode material. In the third stage, the negative electrode of the battery also suffers a certain amount of capacity loss, at the same time, the electrolyte is oxidized, and in the fourth stage of the battery, the battery ruptures. Yuan et al. [25] investigated the temperature difference between inside and outside of a 32 A h pouch battery during overcharging, and found that the internal temperature of the battery was about 140 °C higher than the surface temperature before firing. In addition, it was found that lithium deposition was the main reason for the thermal runaway of the battery. Jiang et al. [26] studied the overcharge behavior and characteristic parameters of cubic lithium ion batteries (43 A h) under different charge rates, conducted risk assessment and early warning analysis, and provided reference for the battery management system. These researches mainly include the change of electrode materials, the behavior and mechanism of battery thermal runaway, and mainly focus on new batteries. However, when the battery is aged, the structure of anode and cathode is destroyed, electrolyte composition is changed and lithium deposition is produced, which changes the performance and thermal stability of the battery. Therefore, the safety of batteries is influenced by their different SOHs [27–29]. Abada et al. [28] combined the model with experimental research and studied the mechanism of LIBs thermal runaway and the

effect of aging on thermal runaway. The results show that calendar aging caused the rise of onset temperature of self-generated heat of the battery, but the trigger of thermal runaway occurred at a lower temperature. Roder et al. [29] conducted a complete study on the relationship between aging and safety of 18650 lithium ion batteries with $\text{LiMn}_2\text{O}_4 + \text{Li}(\text{Ni}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3})\text{O}_2$ composite cathode from the battery level to the material level. In addition, the safety of aging batteries depends on their use history largely [29–34]. Fleischhammer et al. [30] respectively studied the safety of aging batteries under low temperature and high rate, they found that although the cycle under high rate has no great impact on the battery safety, the cycle of aging batteries under low temperature increases the risk of thermal runaway, the stability of SEI films was the main factor influencing the self-reaction onset temperature. However, there are few studies on the thermal safety of aged batteries, especially under dynamic overcharge, and how the heat is accumulated during the overcharge process. In addition, most of aged batteries are obtained by cycling at a low rate, there are few studies on the rapid charging. Therefore, it is necessary to study the characteristics and thermal behaviors of thermal runaway in the dynamic overcharging process of aging caused by rapid charging. The purpose of the experiment reported here was to study, how the thermal behavior and heat accumulation are affected by SOH change caused by rapid charge.

As a current-cutting device is usually installed inside the cylindrical battery, when the overcharge reaches a certain voltage, the current will be cut off and the battery temperature will gradually decrease, without causing thermal runaway. However, the current-cutting device is not used in the pouch batteries, which is more likely to cause thermal runaway due to overcharge. Therefore, 1 A h $\text{LiCoO}_2 + \text{Li}(\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3})\text{O}_2/\text{C} + \text{SiO}_x$ commercial pouch batteries were selected as the research object of this paper. When the SOH is 80%, it is considered to have reached the end of life (EOL) [35]. This paper studied the thermal runaway characteristics of the lithium ion batteries under different SOHs (100%, 90%, 80%) and different charging methods (2 C, 1 C, 0.5C) at the EOL. By monitoring the temperature and voltage in real time, the overcharging behavior and heat accumulation process of the batteries were analyzed, and the relationship between the SOH and the overcharge safety was determined. The purpose of this paper is to provide advice for BMS optimization and battery safety design, at the same time, it also provides data support for preventing the thermal runaway of the battery and the secondary use of the battery.

Table 1

The information of aging cycle tests.

Cycle test	Test procedures	Note
Capacity test	CC (0.2C)-CV charge to 4.2 V until current < 40 mA Rest for 30 min between charge and discharge	Take the average discharge capacity after two cycles
Aging cycle	CC (0.2C) discharge to 2.8 V CC (2 C)-CV charge to 4.5 V until current < 40 mA Rest for 30 min between charge and discharge CC (2 C) discharge to 2.8 V	Cycle ten times

Table 2

The information of all overcharge tests.

Test No.	Initial Capacity (Ah)	SOH (%)	Dynamic overcharge rate (C)
1(1#/2#/3#)	974.85/982.6/963.7	100/100/100	2
2(4#/5#/6#)	981.9/986.7/968.7	91.68/92.3/89.9	2
3(7#/8#/9#)	981.4/987.9/990.45	79.36/82/77	2
4(10#/11#/12#)	984.3/971/980	81.1/83.6/82.6	1
5(13#/14#/15#)	980.9/979/994.3	78.8/83.5/80.5	0.5

2. Experimental

The commercial pouch batteries with $\text{Li}(\text{Ni}_{0.5}\text{Mn}_{0.2}\text{Co}_{0.3})\text{O}_2/\text{LiCoO}_2$ blend cathodes and C/SiO_x anodes with an initial capacity of 1 A h were used. Its voltage range is 2.8–4.2 V, the size of the experimental battery is $52 \text{ mm} \times 34 \text{ mm} \times 5 \text{ mm}$, as shown in Fig. 1. The energy density is 185 Wh kg^{-1} . The melting point of the separator was measured to be 150°C by calorimeter C80 [36,37].

Battery cycle testing consists of two parts, capacity testing and aging cycle, as shown in Table 1. The actual capacity of the battery after cycling is measured by capacity test, and the aging cycle is a way to age the battery. Before the experiment, the capacity test was carried out to obtain the initial capacity of the battery, so as to screen out batteries with similar initial capacity. Between 2.8V and 4.5 V, the capacity test was carried out again every 10 turns to obtain the actual capacity of the battery after aging test. Through the above aging steps, batteries with SOH of 80% and 90% were obtained, that is, the actual capacity accounted for 80% and 90% of the initial capacity respectively.

The prepared experimental batteries were placed in an explosion-proof chamber for further research. The whole experiment process is carried out in the explosion-proof box and recorded by the camera, as shown in Fig. 1. The experimental battery is charged from 0% SOC to overload. The K-type thermocouples are fixed in the center of the upper surface of the battery with high temperature insulating adhesive to record the surface temperature of the battery. The sampling time interval is 0.2 s. The voltage is monitored by the battery test system and the sampling time interval is 1 s.

In our study, overcharge behaviors in different SOHs are analyzed coupled with voltage and surface temperature changes. In order to testify the validation of test results, the overcharge experiment was conducted at least 3 times in each group, there are a total of 15 groups of experiments. An overview of the overcharge tests is given in Table 2. In this table, test number, initial capacity, SOH of the battery, and the current rate used in the process of dynamic overcharge are shown.

3. Results and discussion

3.1. Battery degradation behavior

Fig. 2 shows the aging behavior of the 15# battery. The 15# battery has been cycled for 30 times approximately and reached the end of life (80%SOH). Due to the good reproducibility of the experiment, it can also be used to represent the process of aging to 80% SOH of all batteries in Table 2. The discharge capacity changes of the battery during the whole aging process are drawn in Fig. 2(a). Since this experiment was overcharged, the initial discharge capacity of the battery exceeded 1 A h, which was 1093.1 mA h. However, in the subsequent cycle, it can be seen that the discharge capacity of the battery continues to decay rapidly until it reaches 996.7 mA h. The data points in this figure show a jump change. This is because the actual capacity of the battery needs to be measured at 0.2C after every ten cycles of aging test, so the capacity of the battery at this time has been recovered to a certain extent. Fig. 2(b) is the change of the actual discharge capacity of the battery measured with the number of cycles by capacity test. It can be seen from this figure that the actual capacity of the battery decays in an approximately linear manner. This characteristic is related to the variation of Coulomb efficiency (CE) in Fig. 2(c), and their relationship is explained as follows. In Fig. 2(c), only the CE of the first few cycles is around 100%. With the number of cycles, the CE of the battery gradually drops to less than 1 and is almost stable at 98%. A nearly constant CE below 1 indicates a continuous side reaction in the cell. As a result, the discharge capacity of the battery will decay rapidly. The energy efficiency of the battery is the ratio of the discharge energy to the charge energy, which is relatively low compared with the CE. The relationship between the energy efficiency and the number of cycles can be used to describe the change of the battery impedance during the cycle. The results indicate that side reactions will occur continuously in the process of overcharge test, which will have a certain influence on the thermal behavior and heat accumulation of the battery.

3.2. Overcharge-induced thermal runaway behaviors in different SOHs

The overcharge process of batteries at 2 C rate under different SOHs is shown in Fig. 3. It is obvious that SOH has a great influence on the thermal runaway behavior of the battery. The time of both starting expansion and ignition decrease with the decrease of the SOH. However, generally speaking, the thermal runaway development process of these three SOHs is similar and can be roughly divided into three stages. The first stage is the expansion and deformation of the battery due to the continuous production of gas during overcharge. These gases include CO , CO_2 , CH_4 , C_2H_6 , C_2H_4 and H_2 [38]. The second stage is when the pressure inside the battery increases until it reaches the maximum pressure the battery can tolerate. After that, the gas is ejected from the battery, mixed with air and ignited. The third stage is the process in which the chemical reaction inside the battery ends, the flame goes out, and the battery temperature gradually returns to room temperature. After the end of the experiment, it is found that the batteries are all damaged severely, with the positive and negative sticking together and difficult to distinguish. Silver dots can be seen on each of the charred electrode surfaces, indicating that the temperature inside the battery reaches the melting point of aluminum after thermal runaway. The aluminum current collector is completely oxidized to form silver brittle alumina powder. In addition, in all the overload tests, the nickel bands of the positive electrode melted after thermal runaway occurs, while the nickel bands of the negative electrode are connected, indicating that the reduction of internal heat on the positive side is more intense than on the negative side in the subsequent overcharging experiments.

Due to the similar overcharging behavior of batteries under different life conditions and the good repeatability of each experiment, this paper takes 1#, 4# and 7# batteries as examples to describe in detail the thermal behavior of lithium ion pouch batteries during dynamic

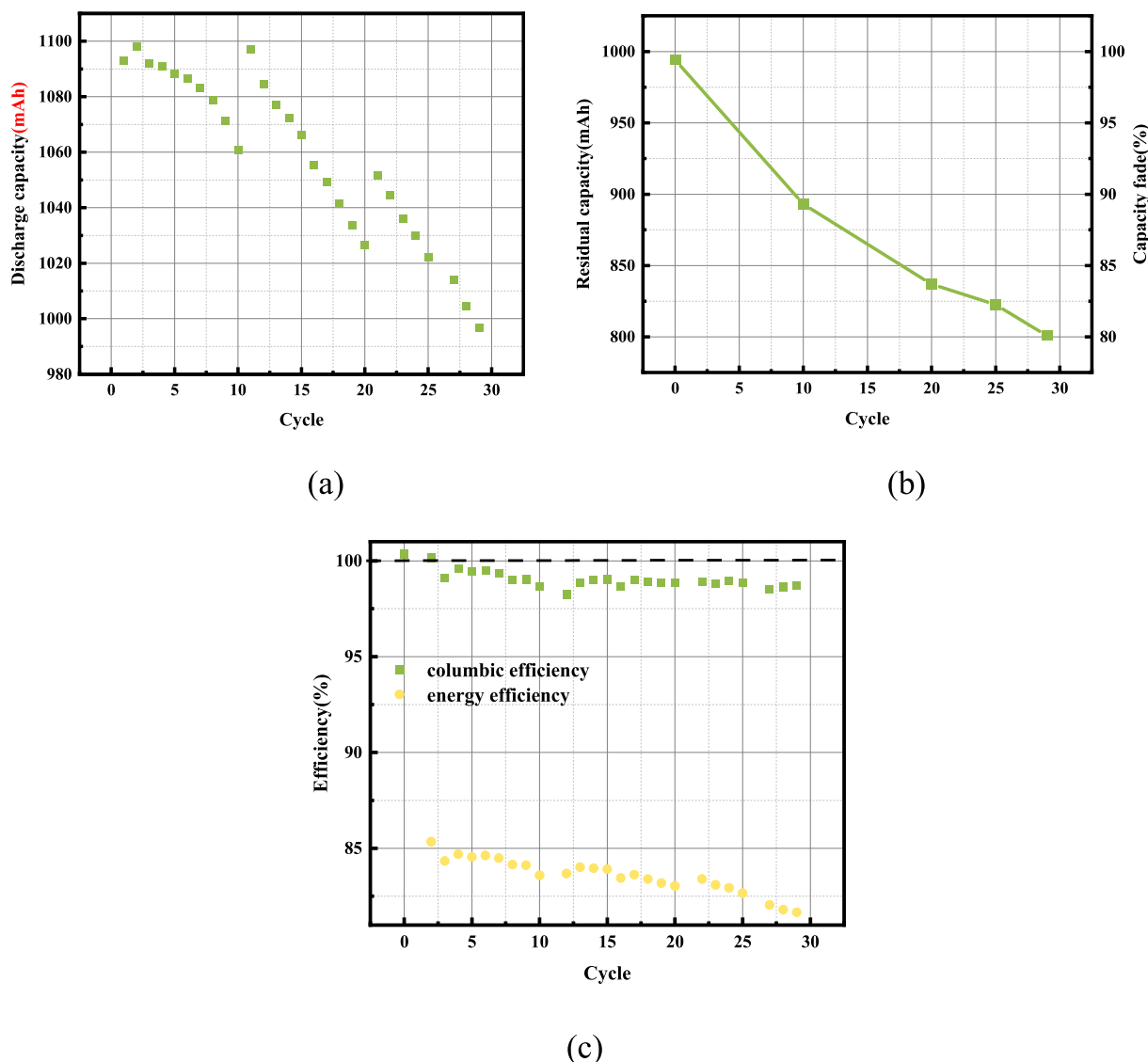


Fig. 2. Batteries aged at 2C until a SOH of 80%, showing discharge capacity in aging test (a), residual capacity in Capacity test (b), the coulombic efficiency and the energy efficiency(b), in dependency of the cycle number.

overcharging. The temperature and voltage changes of 1#, 4# and 7# batteries under 2 C rate are shown in Fig. 4(a)-(c), respectively. According to the change of the three characteristic voltages, the overcharging stage of the battery can be divided into four stages in detail, namely I-IV. This paper takes 80% SOH as an example for detailed description. Then, according to the changes in the four stages under different SOHs, the key parameters in 3.3.2 are extracted and analyzed.

For stage I, in the first 100 s of this stage, the voltage rose rapidly from 3.37 V to 3.95 V, and then rose gently until 4.80 V (171%SOC), and an inflection point (V_{ip}) appeared on the voltage curve. The negative electrode of the battery is designed to have a higher capacity than positive electrode during manufacture process. Therefore, it can continue to be embedded with lithium after it reaches 4.2 V. Then, there is an inflection point on the voltage curve, which is generally considered to be the beginning of lithium deposition in the graphite negative electrode [39]. However, the surface temperature of the battery rose slowly from 30.8 °C to 43.6 °C at this stage, only increasing by 12.8 °C, without great change. The temperature rises at this stage is mainly caused by ohmic heat and irreversible heat generated by electrolyte oxidation [40]. The duration is 2339 s, which accounts for more than half of the overcharge time.

For stage II, when the voltage exceeds V_{ip} , it increases significantly until voltage plateau (V_p) occurs at 5.06 V. After passing this voltage plateau, the voltage will continue to rise until the crest voltage V_c (5.09 V) at this stage. These characteristic voltage value will be discussed later, so it will not be described here. Point A is the temperature point corresponding to the inflection point of voltage (V_{ip}). Compared with Stage 1, the rising rate of temperature at this point increases significantly, which indicates that side reactions occur near the inflection point of the voltage curve [36], such as the reaction between the deposited lithium and electrolyte. In this stage, heat starts to accumulate and the temperature rises faster than the first stage. Point B is the temperature point corresponding to the voltage plateau (V_p), where the temperature rise rate increase. This phenomenon may be due to the fact that the internal temperature of the battery reaches the decomposition temperature of the SEI film [40]. There are two main reasons for this phenomenon. On the one hand, the temperature rise at this stage is more significant than the previous. On the other hand, as the SEI film decomposes, the deposited lithium on the surface of the anode will directly react with electrolyte, which will cause a depletion of lithium deposited on the anode, and then reach a dynamic voltage balance. This process is confirmed by the voltage platform on the curve. The heat generated by

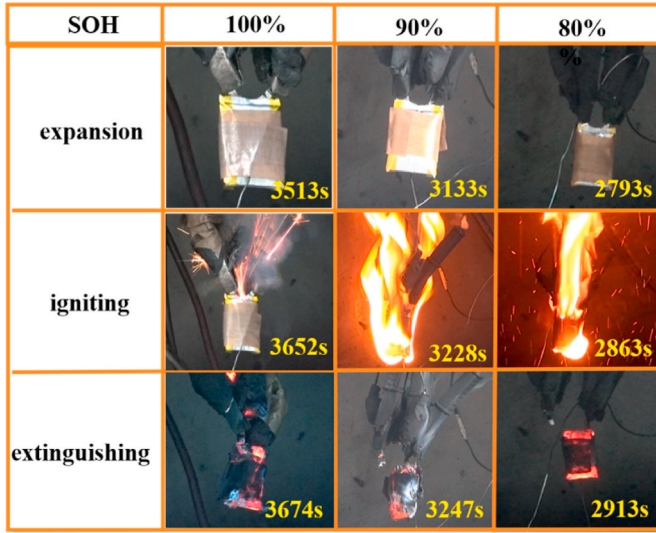


Fig. 3. The overcharging process of the battery at 2 C rate under different SOHs.

this reaction accelerates the temperature rise, and there is a certain temperature difference between the internal and surface of the battery, so the temperature is lower than the decomposition temperature of the SEI film described in the literature (100 °C) [41].

For stage III, due to the continued reaction between the lithium embedded in the anode and the electrolyte, the voltage begins to drop to about 4.84 V [40]. With the continuous accumulation of heat, the pores in separator closed, which rapidly increases the internal impedance of the battery, and the voltage rises to 10.97 V within 22 s. After that, the separator is melted, causing an internal short circuit, and the voltage drops to 0 V rapidly, triggering the thermal runaway. In addition, it was found that the volume of the battery began to expand at this stage, and the battery expand was usually caused by the decomposition of electrolyte and SEI film.

For stage IV, at this stage, thermal runaway is triggered, and a series of exothermic reactions occur, including electrolyte oxidation, SEI film decomposition, the reaction between cathode and the electrolyte, binder decomposition, electrolyte and lithium deposition, and internal short circuit after separator melting, etc. These reactions will produce a large number of gases, including CO₂, CO, CH₄, C₂H₆, C₂H₄ and H₂, etc [38]. The gas is ejected from the broken battery and ignited at high temperatures, rising rapidly from 103.2 °C to 681.8 °C in seconds. Because the charging current at this stage drops to 0 A, it is impossible to continue overcharging. When all the chemical reactions inside the battery stop, the internal temperature of the battery gradually drops to room temperature, and the internal system of the battery rebalances.

3.3. The effect of aging on dynamic overcharge

3.3.1. Heat accumulation at different SOHs

According to Eq. (1) and temperature rise rate data, the heat accumulation value of the battery at any moment during the whole charging process can be obtained. Where, Q_h represents the total heat accumulated during charging, and the unit is J, m is the mass of the battery, which is 17.3 g, c is the specific heat capacity, which has been measured in Ref. [34], so it will not be repeated here, is $1.0 \text{ J g}^{-1} \text{ K}^{-1}$ in this paper [34].

$$Q_h = \int \frac{dQ}{dt} dt = \int \frac{cmdT}{dt} dt = cm \int \frac{dT}{dt} dt \quad (1)$$

Therefore, by integrating the temperature rise rate of 1#, 4# and 7# batteries, the heat accumulation of these batteries in any time period can

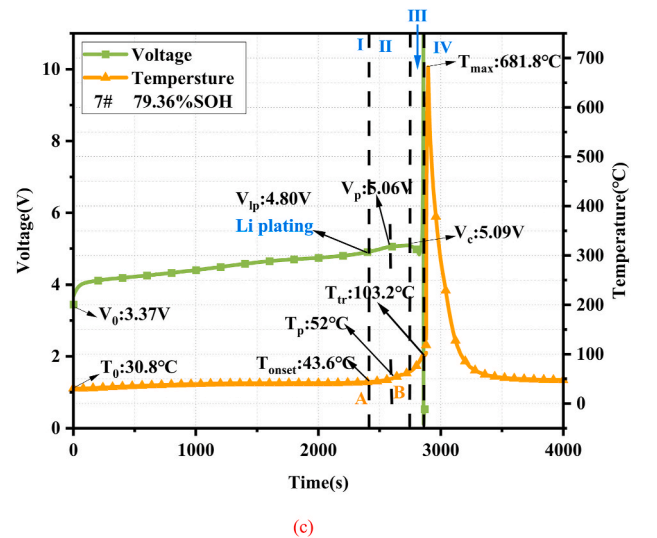
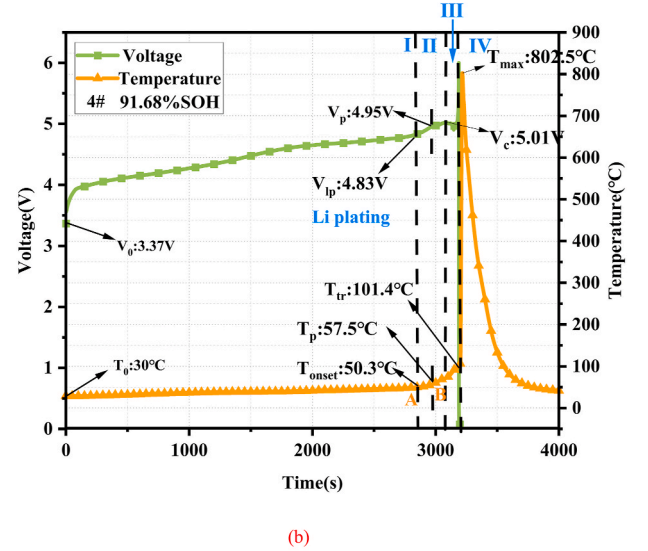
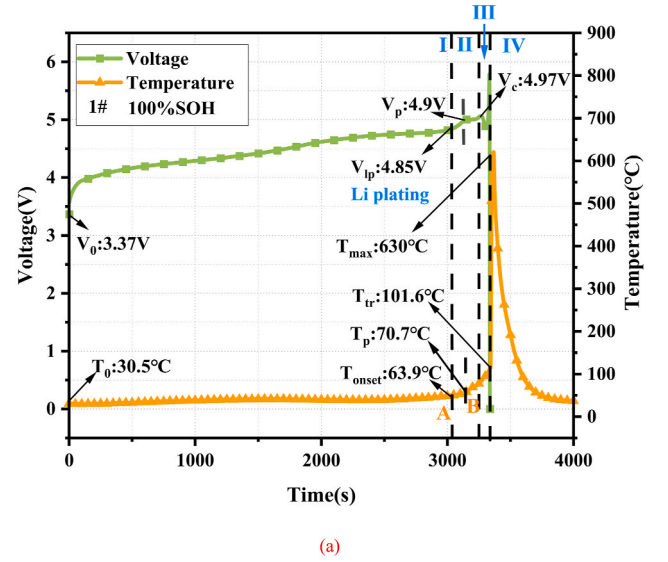


Fig. 4. Voltage and surface temperature curves with different SOH (a: 1#, 100% SOH; b: 4#, 91.68% SOH; c: 7#, 79.36% SOH).

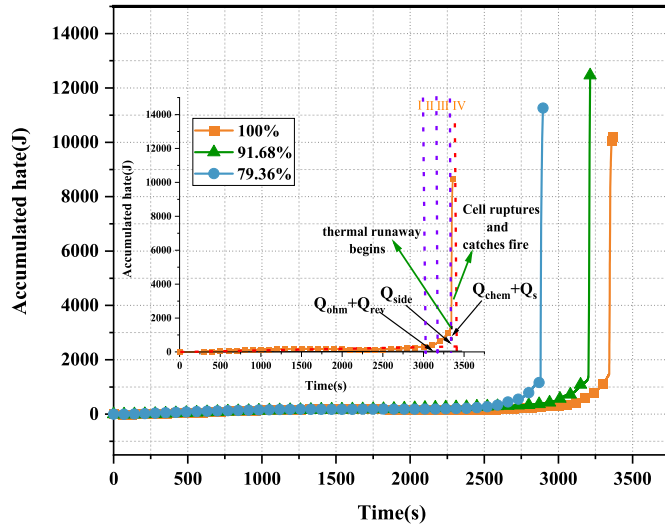


Fig. 5. The accumulation of heat in the battery during overcharge.

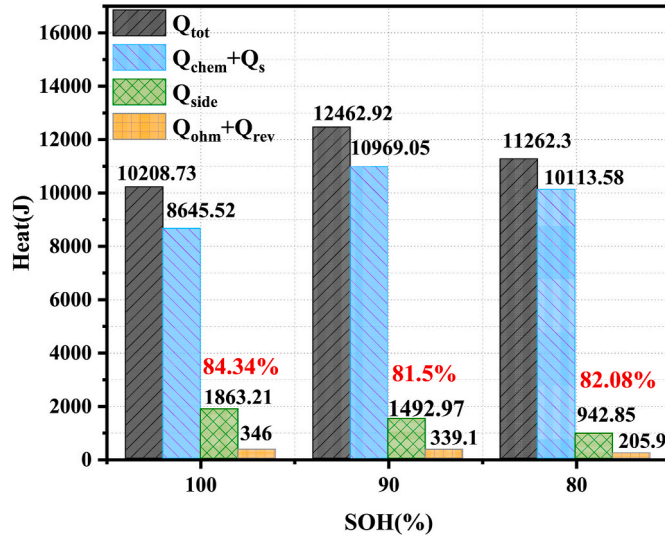


Fig. 6. The contribution rate of side reactions under different SOHs to battery thermal runaway.

be obtained, as shown in Fig. 5. The illustration in Fig. 5 shows the heat accumulation in the process of overcharging according to Ref. [37] and Eq. (2), including the heat accumulation of Q_{side} and $Q_{ohm} + Q_{rev}$ before thermal runaway and $Q_{chem} + Q_s$ after thermal runaway. Q_{tot} is the heat generated during the entire charging process, Q_{side} is the heat accumulation caused by the side reaction inside the battery before the battery thermal runaway occurs, Q_{ohm} is the joule heat caused by the internal impedance of the battery, Q_{rev} is the reversible heat of the battery, Q_s represents the heat released by the internal short circuit, and Q_{chem} represents the accumulation of heat generated by all the internal chemical reactions inside the battery after the thermal runaway occurs.

$$Q_{tot} = Q_{side} + Q_{ohm} + Q_{rev} + Q_s + Q_{chem} \quad (2)$$

Table 3

The ratio of heat accumulation and total electric energy from initial state to V_{lp} , V_c and V_{end} under different SOHs.

SOH (%)	Q_{h1} (J)	Q_{e1} (J)	η_1 (%)	Q_{h2} (J)	Q_{e2} (J)	η_2 (%)	Q_{h3} (J)	Q_{e2} (J)	η_3 (%)
100	346	26286.6	1.31	875.38	29226.9	3.00	1416.87	29796.2	4.76
91.68	339.1	26084.6	1.30	707.57	27408.3	2.58	1273.28	28521.2	4.46
79.36	205.9	21503.3	0.96	560.52	24789.6	2.26	1159.1	26081.1	4.44

Table 4

Characteristic parameters of overcharge test under different SOHs.

Battery	SOH (%)	V_{lp} (V)	V_p (V)	V_c (V)	T_{onset} (°C)	T_{tr} (°C)	T_{max} (°C)
1#	100%	4.85	4.90	4.97	63.9	101.6	630
4#	91.68%	4.83	4.95	5.01	50.3	101.4	802.5
7#	79.36%	4.80	5.06	5.09	43.6	103.2	681.8

The heat accumulation of the battery during charging is shown in the bar chart in Fig. 6, which can be intuitively seen that the heat of $Q_{chem} + Q_s$ part increases after battery aging, but Q_{side} gradually decreases with the decrease of SOH, and the Q_{side} of the new battery is twice that of the battery at the end of life.

Q_{side} can be ignored when the battery is used under normal conditions, but when the battery is used under abusive conditions such as overcharge, the heat generation of this part cannot be ignored. During the period after V_{lp} to before thermal runaway, as the heat accumulation of the battery only consists of Q_{side} , Q_{ohm} and Q_{rev} , the contribution rate of side reactions to thermal runaway at different life stages can be calculated by referring to Eq. (3), and the results are also shown in Fig. 6. It can be seen that although the heat generated by side reactions is different in different SOHs, its contribution rate to thermal runaway is almost not affected by the degree of battery aging, and both provide more than 80% of the heat, and the occurrence of thermal runaway is mainly dominated by side reactions. Among them, the oxidation reaction of electrolyte and the reaction of lithium ion with electrolyte contributed the most to the thermal runaway [23]. A large number of lithium deposited on the negative electrode reacts with the electrolyte, resulting in a significant rise in battery temperature, which eventually leads to the melting of the separator, triggering a series of reactions such as the decomposition of the positive electrode, exothermic reaction between the positive electrode and the electrolyte, and finally leading to the irreversible thermal runaway of the battery [41,42].

$$\xi = \frac{Q_{side}}{Q_{tot}} = \frac{Q_{side}}{Q_{ohm} + Q_{rev} + Q_{side}} \quad (3)$$

Meanwhile, the input electric energy can be obtained by calculating the integral of UI with respect to t by Eq. (4). Where, Q_e is the amount of electricity that is charged into the battery. When the battery voltage reaches V_{lp} , the accumulated heat is Q_{h1} and the charged electric energy is Q_{e1} . When the voltage reaches V_c , the accumulated heat is Q_{h2} and the charged electric energy is Q_{e2} . When the charging ends, the accumulated heat is Q_{h3} and the charged electric energy is Q_{e3} . The electric energy conversion efficiency η is calculated by the ratio of the accumulative heat to the power and recorded in Table 3. The greater the value, the lower the electric energy conversion efficiency of the battery. After the aging of the battery, although the change range of values η_1 and η_3 is small, the three values all decrease, which indicates that the energy conversion efficiency of the aging battery is higher than that of the new battery. In addition, it can be seen that with the progress of charging process, the heat accumulation rate keeps increasing, and the energy conversion efficiency gradually decreases.

$$Q_e = UIt \quad (4)$$

3.3.2. Key parameters change under different SOHs

In order to make it more convenient to compare the changes of various characteristic parameters, these parameters involved in

Table 5

The duration of the overcharge tests with different SOHs.

Battery	SOH	t_1	t_2	t_3	t_{tot}	t_2+t_3
1#	100%	3020	223	101	3344	324
4#	91.68%	2806	268	130	3204	398
7#	79.36%	2339	413	116	2868	529

overcharging at different SOHs are listed in Table 4. It can be seen that the onset temperature of exothermic reaction (T_{onset}) and the voltage when the reaction of lithium deposition takes place (V_{lp}) decreases, which indicates that battery becomes more dangerous after aging.

V_{lp} is a turning point on the voltage curve. As we know, this point indicates that the graphite negative electrode begins to deposit lithium [34]. As can be seen from Table 4, V_{lp} decreases with the decrease of SOH. This is due to the loss of active lithium ions to a certain extent in the high-rate overcharge cycle of the battery. Therefore, for a new battery, more lithium will be embedded in the graphite negative electrode before the inflection point of the voltage curve, resulting in a lower negative potential than the aging battery. In addition, the temperature after this point was found to rise more quickly than in the previous stage, indicating a series of side reactions taking place inside the cell. Since the abnormality of this point can be easily found from the voltage curve in the early stage, this voltage value can be used as an indicator of battery failure.

V_p is a platform on the voltage curve that corresponds exactly to the inflection point on the temperature curve. The reason for this platform is that the internal temperature of the battery reaches the decomposition temperature of the SEI film, and with the decomposition of SEI film, the lithium deposited on the anode surface will react directly with the electrolyte, consume part of the lithium deposited on the anode, and then reach a dynamic balance with the lithium deposited on the anode [40]. This value is close to V_c and generally speaking, it is not affected by the SOHs.

T_{onset} corresponds exactly to the inflection point on the voltage curve, where the temperature starts to rise gradually from the first stage, indicating the beginning of some side reactions. As shown in Table 4, T_{onset} of batteries with 80%, 90% and 100% SOH are 43.6 °C, 50.3 °C and 63.9 °C respectively. Therefore, the temperature at this point is related to the aging state of the battery, and generally decreases with the decrease of SOH. In other words, side reactions occur at lower temperatures. This may be due to the fact that, as the battery ages, the rising

rate of voltage increases and therefore the duration of the Stage I is reduced, so the heat accumulated by ohmic heat and electrolyte oxidation during this stage is correspondingly reduced.

T_{tr} refers to the initial temperature when the thermal runaway is triggered. As can be seen from Fig. 4, this temperature point corresponds to the sudden drop of the battery voltage. As shown in Table 4, the temperature value does not change with the battery SOH and is almost stable at 103 °C. When the temperature inside the battery reaches the melting temperature of the separator, it will cause an internal short circuit and trigger the thermal runaway, which presents a sudden drop in voltage. The temperature barely changed, which means the separator is unaffected by the aging process of battery. However, in this stage, the time of thermal runaway occurs in advance with the decrease of SOH. As the separator began to melt, the internal temperature of the battery was close to 150 °C, but the measured surface temperature was about 100 °C, indicating that the temperature difference between the inside and outside of the battery was about 50 °C.

T_{max} is the maximum temperature on the surface of battery, which could describe the maximum amount of heat that battery releases after thermal runaway. Many investigators have explored the relationship between SOH and T_{max} . Different scholars have put forward different views. Some studies show that, as SOH increases, its influence on T_{max} decreases. This is because the energy stored in battery decreases the aging process deepens [27,40]. In the process of experimental research of this work, T_{max} could not be measured accurately because the connection between the thermocouple and the surface of battery may become loose. Therefore, the relationship between T_{max} and SOH could not be explored accurately.

3.3.3. The duration of different stages of overcharge process under different SOHs

Table 5 lists the duration of each stage of the battery in different SOHs before thermal runaway, where t_1 , t_2 and t_3 represent the duration of corresponding stages respectively, t_{tot} represents the total duration before the thermal runaway, which is the sum of t_1 , t_2 and t_3 .

Fig. 7(a) is the duration of each phase of the battery under different SOH before thermal runaway, and Fig. 7(b) is an enlarged view of the part of Fig. 7(a), involving t_2 , t_3 and t_2+t_3 . As can be seen from Fig. 7(a), t_1 , t_2 and t_{tot} just have a linear relationship with SOH, and can be expressed by Eqs. (4)–(6), and their R-square is 0.95602, 0.91547 and 0.94651. t_3 almost does not change with SOH.

After the battery aging, the duration of stage I is significantly

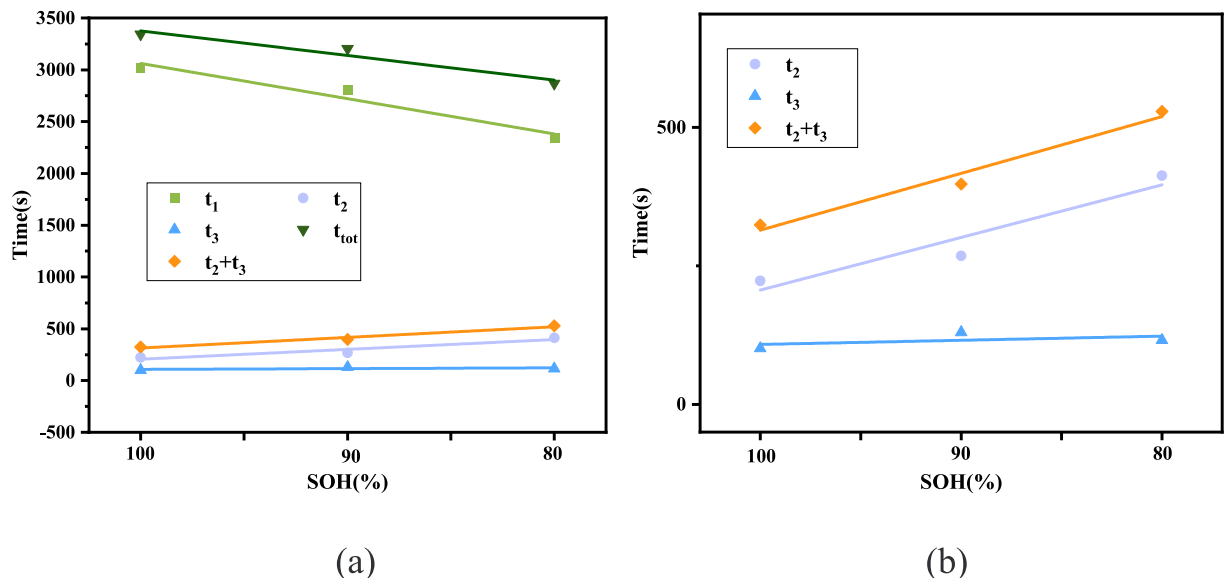


Fig. 7. The duration of Stage I, II and III under different SOH.(b is an enlarged view of part of a).

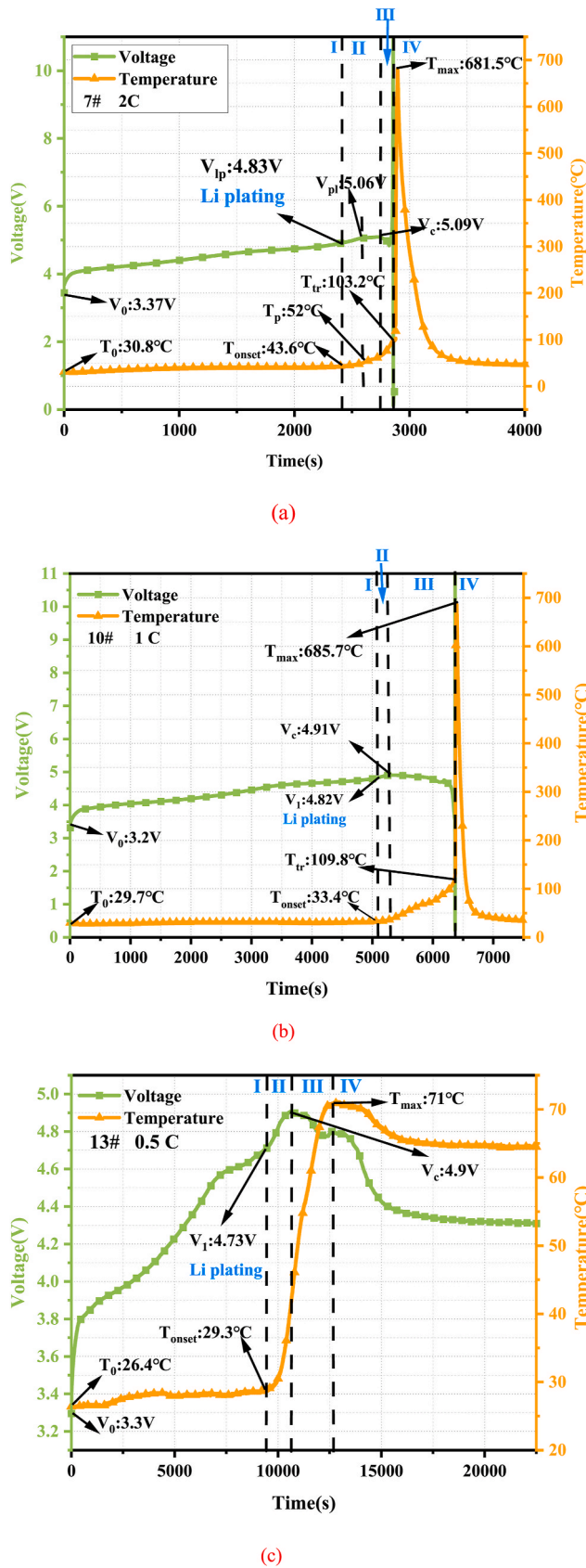


Fig. 8. Voltage and surface temperature curves with different C-rates(a: 7#, 2C; b: 10#, 1C; c: 13#, 0.5C).

shortened, mainly because the negative electrode capacity of the battery has a certain degree of degeneration after aging, which makes it easier for the battery to reach the maximum capacity of the negative electrode, and thus lithium deposition will occur earlier than the new battery. In addition, with the aging of the battery, the internal active materials decrease and the activity of the material decreases, so the t_2 gradually increases with the decrease of SOH. Although t_2 increases with aging, when the battery is aging, the total duration before thermal runaway decreases sharply with SOH, and the battery safety decreases.

As mentioned above, the occurrence of thermal runaway can be warned in advance by the emergence of characteristic voltage V_{lp} , so the period from V_{lp} to thermal runaway, namely $t_2 + t_3$ is analyzed. Since the duration of stage III changes little with SOH, the duration of $t_2 + t_3$ from V_{lp} to thermal runaway is mainly related to stage II. For the new battery, the duration of this stage is 324 s. When the SOH is 90%, the duration is about 398s. When the SOH is 80%, the duration is about 529 s. In other words, when the battery ages, there is more time to warn of the risk of thermal runaway. Therefore, this value provides a feasible way to forewarn the imminent risk of thermal escape, that is, when the voltage exceeds the V_{lp} , effective safety measures (such as cutting off the current and starting the cooling system) must be taken within 3 min to prevent thermal runaway of the battery.

$$t_{tot} = 2380SOH + 996.67 \quad (5)$$

$$t_1 = 3405SOH - 342.83 \quad (6)$$

$$t_2 = -950SOH + 1156.3 \quad (7)$$

3.4. Influence of overcharging method on aged batteries

The previous analysis showed that the safety was lowest when the battery SOH was 80%. We know that the joule heat is proportional to the current, and when the charging current increases, the heat production significantly increases, so the battery risk will also significantly increase. In order to study the influence of the charging rate on the thermal behavior during dynamic overcharge, the batteries with 80% SOH aging were selected. The purpose is to select the most extreme case to study the impact of charging mode on aging batteries. Due to the good repeatability between experiments of each group, the 7#, 10# and 13# battery were taken as examples to describe the influence of overcharging mode on aged batteries in detail, as shown in Fig. 8. As expected, the rate has a great influence on the thermal runaway behavior. With the increase of the charging rate, the duration of the entire charging process is shortened, and the time of side reaction and the onset time of thermal runaway also decrease.

When the charging rate was 1 C, thermal runaway occurred, which was similar to the thermal behavior under 2 C. The triggering of thermal runaway corresponds to the voltage dip, which is further explained that the cause of thermal runaway of the battery is triggered by the separator melting and does not change with the charging rate. The voltage of lithium deposition does not change much, which is due to the fact that more capacity is charged into the battery at low rate, making the negative potential lower while the polarization potential smaller at low rate [40]. To our surprise, T_{max} doesn't increase with the decrease in the charge rate, although more energy was charged at the low rate. Moreover, T_{onset} also decreases with the decrease of charge rate.

When the charging rate is 0.5C, the heat generation rate of the side reaction inside the battery is relatively low, and there is enough time for heat exchange with the surrounding environment. Therefore, when the battery is overcharged at 0.5C, thermal runaway would not happen but the serious expansion is observed. Voltage first rises then falls after reaching the crest voltage, which is correspond to the lithium in the negative electrode and electrolyte continue to react to produce metastable SEI films in the subsequent reaction process, which will form a certain protective layer between the positive and negative electrodes.

However, metastable SEI films will decompose at high temperature. To sum up, the higher the charge rate, the worse the thermal safety of the battery.

4. Conclusions

In this work, by detecting the voltage and surface temperature changes of $\text{LiCoO}_2 + \text{Li}(\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3})\text{O}_2/\text{C} + \text{SiO}_x$, the dynamic overcharge behavior when the battery is rapidly overcharged to different SOHs is studied in detail. The thermal safety of the battery has a strong correlation with the SOH. Aging batteries are more dangerous. This is mainly because the loss of lithium ions leads to earlier lithium deposition in the aging batteries, and the loss of active anode materials reduces the value of t_1 . In addition, batteries charged at high rate are more dangerous.

According to the voltage change, the overcharge process can be divided into four stages. The continuous accumulation of heat causes the melt of separator, which would lead to an internal short circuit, and trigger thermal runaway eventually. The temperature stabilizes at approximately 103 °C, and is not affected by the SOH. After V_{ip} , side reactions play an important role in the thermal runaway of the battery. Under the charging rate of 2 C, these side reactions provide more than 80% of the heat for the thermal runaway, which are not affected by the battery SOH. Therefore, effective control of lithium deposition and reduction of reaction heat between electrolyte and lithium deposition are the keys to improve battery safety. From the perspective of safety, V_{ip} provides a feasible method to predict the possible risk of thermal runaway. When the charging rate is higher than 1 C, once the voltage value is exceeded V_{ip} , effective cooling methods must be adopted within 3 min. If the battery cannot be cooled in time, once the temperature reaches T_{tr} , fire extinguishing equipment should be worked as soon as possible. In addition, the energy conversion efficiency decreases gradually in the whole process, but increases with the decrease of SOH, except at the end of charging.

The results of this study can better understand the heat accumulation process and the contribution of side reactions during overcharging under different SOHs. In addition, it provides guidance for battery safety design and safety monitoring of BMS. The experimental battery used in this paper was obtained through cycle aging, but the aging condition could not fully reflect the actual aging process. The future work will focus on the impact of different aging histories on overcharge behavior.

CRediT authorship contribution statement

Lei Feng: Conceptualization, Investigation, Methodology, Data curation, Writing – original draft. **Lihua Jiang:** Validation. **Jialong Liu:** Validation. **Zhaoyu Wang:** Methodology. **Zesen Wei:** Validation. **Qingsong Wang:** Supervision, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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